

Empowering Human Learning through Human-Robot Interaction

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Abstract

While many robots are designed to assist people, their increasing intelligence across social and cognitive tasks may hinder humans from developing their own abilities due to over-reliance. We present prior, ongoing, and proposed work to explore how human-robot interaction (HRI) can instead empower human learning. First, we examined how robots integrate into learning ecosystems, uncovering that learning supporters prefer robots to act as their collaborators instead of replacement. Our framework identified collaboration mechanisms and was validated through prototype development and field evaluation. Next, we investigated how mobile robots can enhance in situ learning in the physical world through situated cognitive mechanisms. Finally, we plan to develop robotic arm prototypes that convey abstract learning concepts through manipulation of physical objects. Through these efforts, we aim to transform HRI from assistance to empowerment, enabling humans to enhance their capabilities through meaningful interactions.

CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; • **Applied computing** → Interactive learning environments.

Keywords

Human-Robot Interaction, Human Learning, Ecosystem Integration, In Situ Discovery, Physical Demonstration

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1 Introduction

Many robots are designed to *help* people in various ways—from automating repetitive tasks in industrial settings [1, 15, 25, 30] to supporting the daily routines of older adults in caregiving environments [9, 19, 27, 29]. With the rapid advancement of artificial intelligence (AI) and large language models (LLMs), interactive systems, including robots, are becoming increasingly capable of engaging in cognitively complex conversations and interactions [6, 8, 16, 17, 31]. Examples include learning companion robots that

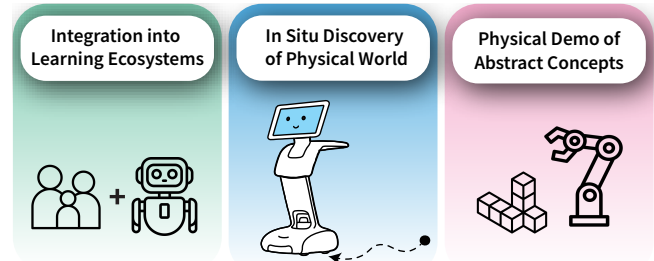


Figure 1: We unfold three layers of mechanisms for empowering human learning through HRI, including integration into learning ecosystems through social mechanisms (*left*), in situ discovery of physical world through cognitive mechanisms (*middle*), and physical demonstration of concepts through tangible mechanisms (*right*).

assist children’s education [5, 10], tour guide robots that facilitate knowledge and information access [4, 14, 22], and emotional support robots that provide outlets for sharing personal feelings [7, 18]. Prior work has highlighted potential issues related to human reliance on robots when it comes to developing and maintaining human’s own capabilities [12, 23, 28]. This prompts a critical question: *How can everyday human–robot interaction be leveraged to enhance rather than erode human capability development?*

Existing related work has demonstrated how several properties of robots—such as physical embodiment, social interaction, and physical co-location—can benefit learning, compared to traditional educational technologies [2, 3, 20, 21, 24, 26]. However, most studies have focused on controlled, task-specific contexts—such as tutoring sessions or problem-solving tasks—typically involving one-on-one interaction between a learner and a robot. While these designs yield insights into HRI mechanisms for learning facilitation, they do not yet address how HRI can be meaningfully integrated into *everyday life*, where learning often occur incidentally. The challenge, then, lies in designing HRI that can naturally coexist with human routines and subtly scaffold human growth without displacing human agency that drive intrinsic learning.

In this work, we discuss our prior, ongoing, and future work for pushing the boundaries of HRI by rethinking how these interactions can go beyond simply *helping* people to complete predefined tasks towards *empowering* individuals to develop and apply their own capabilities in real-world contexts. Based on our prior work [10, 13], we unfold the potential aspects in which physical robots can uniquely empower human learning experiences: *collaborative learning support ecosystem*, *in situ learning in physical world*, and *physical/tangible demonstration of abstract concepts*. Throughout the work, we ask the following research questions:



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- (Social) **RQ1**: How can robots be integrated into human learning support ecosystems?
- (Cognitive) **RQ2**: How can robots enhance in situ learning in the physical world through situated cognitive mechanisms?
- (Tangible) **RQ3**: How can robots demonstrate learning concepts through manipulation of physical everyday objects?

2 Integration into Learning Support Ecosystem

In our *prior* work, we conducted two studies to investigate how robots can be integrated into existing learning support ecosystems: (1) We proposed a systematic framework for collaboration mechanisms among learning supporters and robots through a technology probe study. (2) We built prototypes operationalizing the framework and tested its usage patterns in the field.

Collaboration Mechanism Framework for Supporting Learning [11]. In this work, we examined how robots can be designed to support interactions within parent–child dyads, recognizing the vital role of parental involvement in children’s learning. Building on insights from Ho et al. [13], we used a technology probe (i.e., a prototype robot capable of autonomously delivering educational prompts and responses during reading) to explore how robots can be integrated into family learning contexts ($n = 10$). We found that parents preferred to *collaborate* with the robot to jointly support their child’s learning. Based on our data, we introduced the CAM framework—capability (C), availability (A), and motivation (M)—to model when and what responsibilities should be shared between parents and robots, illustrating how robots can be integrated into learning ecosystems through stakeholder collaboration [11].

Develop Prototype for Flexible Involvement in Learning Support [12]. In this work, we developed design methods and prototypes to operationalize the framework proposed in [11]. We first created a card-based activity to systematically map various scenarios of parental involvement. Then, we implemented an LLM-assisted interface integrated with a robot, allowing parents to review and revise LLM-generated learning content while dynamically adjusting their level of involvement in the learning activity by delegating roles between themselves and the robot. We conducted an in-home study with parent–child pairs ($n = 20$; children aged 3–5 years) to examine parental involvement scenarios, use of LLM-generated content, and collaboration patterns with the robot. Our findings reveal distinct collaboration patterns between parents and robots across different scenarios. For instance, when parents are occupied but confident in the task, they opt for an HRI mode where the robot assumes full responsibility, allowing for autonomous support. Conversely, when parents are available but less confident in their pedagogical abilities, they prefer a shared-responsibility mode where the robot can provide support while keeping themselves involved. These patterns highlight the need for HRI designs that empower parents to *calibrate their level and type of involvement dynamically*. Furthermore, we found that *task difficulty should be synchronized with these interaction dynamics to maximize the value of parental presence*: increasing complexity when parents are present allows them to scaffold complex concepts and mitigate frustration, while simplifying tasks during solo child-robot interactions ensures the child remains engaged without needing parental support.

3 In Situ Discovery of Physical World

Besides collaborating with human through flexible autonomy to support learning ecosystems, in this *ongoing* work we introduce another pillar of strategy to enhance human learning through HRI. Based on situated cognition, we argue that the unique mobility and agency of robots can support human in situ learning by offering verbal conversations about learners’ situated environment. To investigate this, we conduct a two-phase study involving adults and children (aged 3–10 years): Phase one involved an in-home co-design study ($n = 12$) using a low-fidelity prototype to uncover design considerations. Phase two involved the development of a semi-autonomous prototype, RobAi, capable of capturing environmental scenes and generating learning-oriented conversations, followed by an in-home study ($n = 15$) to observe natural HRI patterns.

We uncovered design dimensions of mobile robots for in situ learning based on interview data and interaction patterns. Our findings suggest several design implications: (1) *Augmentation and Orientation*: prioritize augmenting learner’s cognition while proactively orienting their attention to overlooked environmental elements; (2) *User-controlled Autonomy*: allow users to modulate agency across initiation (proactive vs. reactive), interaction dynamics (single vs. multi-user), and navigation (leading vs. following); (3) *Physical Safety and Social Norms*: override unsafe MLLM suggestions, enforce social and privacy boundaries (e.g., avoiding bathrooms), and report sensitive-topic interactions to guardians; (4) *Age-adaptive Interaction*: employ age-adaptive strategies, such as redundant multi-modal inputs (verbal, touch, and gesture), to empower younger children’s autonomy, mitigate dominance in group settings, and reduce parental dependency; (5) *Ownership and Deployment*: optimize personal robots for portable affordability and individual privacy, while engineering public robots for multi-party interaction and collective data security.

4 Physical Demonstration of Abstract Concepts

Extending beyond verbal conversations about in situ scenes in the physical world, our proposed *future* work aims to investigate how robots with manipulation capabilities can manipulate physical objects to explain concepts, extending in situ learning from conversational-based to tangible and embodied experiences. Robotic arms can manipulate physical objects or perform dynamic visualizations that demonstrate or metaphorically represent cognitive concepts. These concepts may span a wide spectrum—from early educational topics such as basic mathematics (e.g., manipulate blocks to explain addition or subtraction) to more advanced domains such as the structure of the solar system (e.g., use objects to represent planets and demonstrate the sequence and distance among planets). Crucially, unlike mixed-reality or video-based demonstrations, robotic arms can physically interact with learners. They can scaffold understanding by collaborating with learners on tasks tangibly—e.g., placing one block on the left and two blocks in the middle, then prompting a child to add the corresponding number of blocks to represent the sum on the right. This work will demonstrate how robotic manipulation applied to physical objects can create meaningful and interactive learning opportunities.

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